

Matrix Factorization Recommender System: Preprocessing, Analysis, and SVD Modeling

1 - Executive Summary

This report presents the complete workflow for building a collaborative filtering movie recommender system using SVD on the large-scale **ml-32m** dataset. After merging and cleaning the raw files, the final dataset contained **32M user-item interactions** with engineered features for genres and timestamps.

EDA showed a strong **positive rating bias** (4.0 most common) and a **stable taste profile** across nearly three decades, despite fluctuations in yearly activity.

The final SVD model was trained using an **80/20 train-test split**. It achieved strong accuracy with **RMSE = 0.8259** and **MAE = 0.6243**, and delivered effective recommendations with **Precision@10 = 0.6625** and **Recall@10 = 0.6679**. Additional experiments confirmed that model performance improves significantly with larger training data.

2 - Data Foundation: Preprocessing and Feature Engineering

2.1 Initial Data Sourcing and Assembly

Four raw files were imported from the ml-32m dataset: **links.csv**, **movies.csv**, **ratings.csv**, and **tags.csv**. To build the user-item interaction matrix needed for collaborative filtering, **movies** was merged with **ratings** using a left join on *movieId*. This produced a consolidated dataframe **df** with **32M+ rows** and **6 initial columns**.

2.2 Data Cleaning

Post-merge checks revealed **3,153 nulls** in *userId*, *rating*, and *timestamp*—all originating from movies with **no ratings**. Since unrated movies provide no value in a collaborative filtering setup, these rows were removed using **df.dropna()**. The resulting dataset contained **32,000,204 fully clean entries**, with **zero remaining nulls**.

2.3 Feature Engineering I: Transformation of genres Column

The raw genres column was identified as a critical feature for content analysis, but its format—a single string with genres separated by a pipe (|) delimiter (e.g., "Adventure|Animation|Children")—was unsuitable for machine learning or quantitative analysis.

To structuralize this data, a one-hot encoding process was designed and implemented.¹ First, the genres string in each row was split by the | delimiter. From the complete set of all genres in the

dataset, a new binary column was created for each unique genre. A 1 was populated if the movie belonged to that genre and a 0 otherwise.

During this process, a specific genre entry, (no genres listed), was identified. This was renamed to a more descriptive and syntactically-valid column name, Unknown. This entire transformation expanded the dataframe by 20 new feature columns (from Musical to Unknown), enabling granular, content-based exploratory analysis.

2.4 Feature Engineering II: Processing of timestamp Column

The raw timestamp column was provided in Unix format (i.e., seconds since the epoch). To unlock this feature, the column was converted into a standard datetime object using the `pd.to_datetime(df['timestamp'], unit='s')` function. A new feature, `rating_year`, was engineered by extracting the year component from the new datetime object. This `rating_year` column proved essential for the temporal analyses detailed in the EDA section, allowing for the examination of user activity and rating behavior over time.

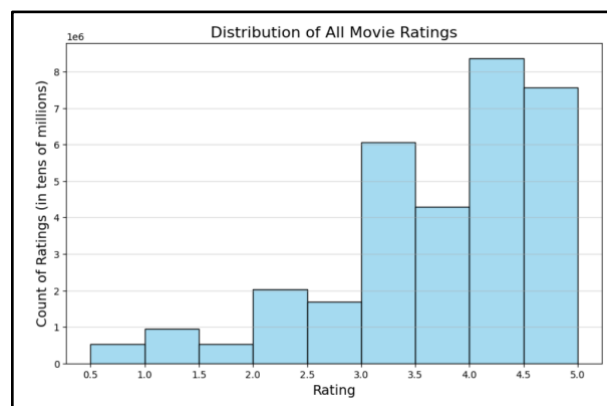
2.5 Final Analytic Dataset

After merging, cleaning, and feature engineering, the final dataset contained **32,000,204 rows** and **26 columns**. It was saved as `cleaned_movies.csv`, serving as the consistent, reproducible dataset for all downstream analysis and modeling.

3 - Exploratory Data Analysis (EDA) of the Ratings Landscape

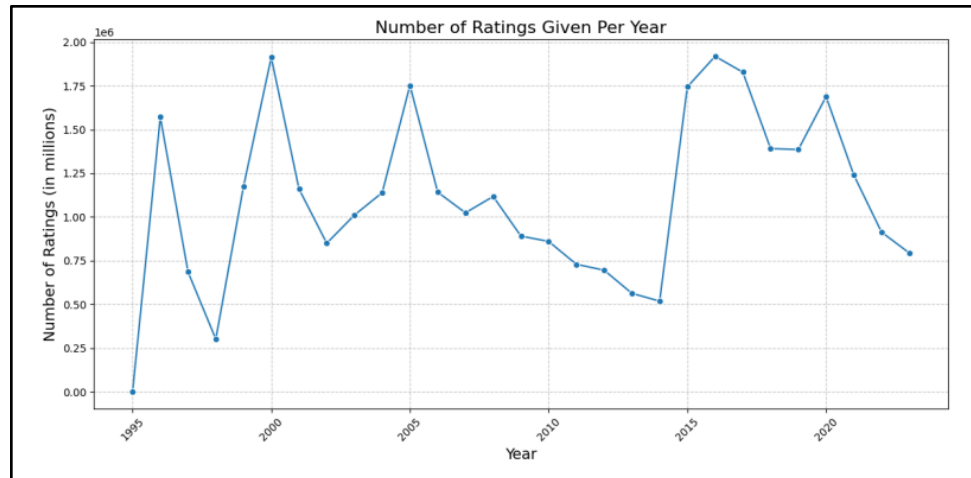
3.1 Distribution of All Movie Ratings

A histogram of all 32M ratings showed a strong positivity bias: **4.0** is the most common rating, followed by **4.5** and **5.0**, while low scores (0.5–1.5) are rare. This indicates users primarily rate movies they enjoy, meaning the dataset contains far more “positive” signals than negative ones.



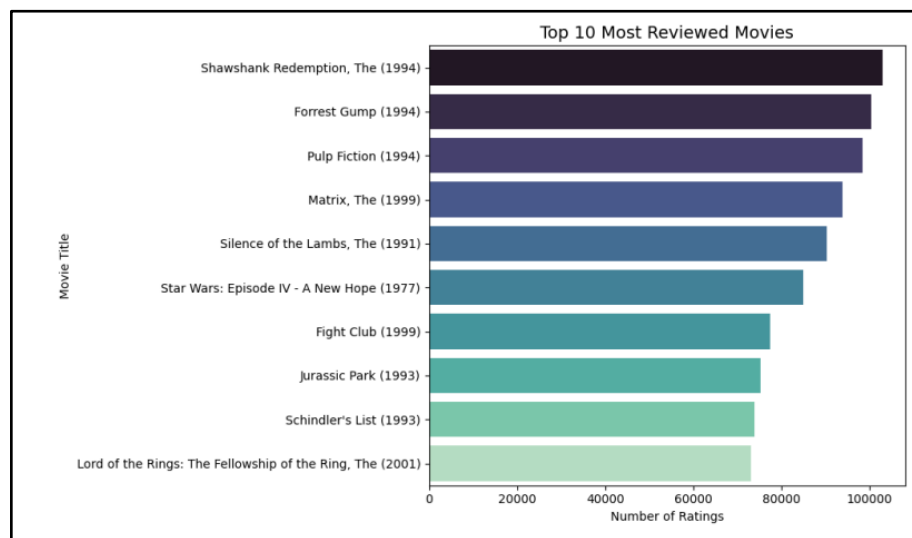
3.2 Temporal Analysis: Platform Activity

Yearly rating counts (1995–2023) showed rapid early growth, several peaks around **2000**, **2005**, and a major surge between **2015–2018**. Activity fluctuates significantly over time, reflecting a non-linear and highly dynamic user base.



3.3 Content Popularity: Top 10 Most-Reviewed Films

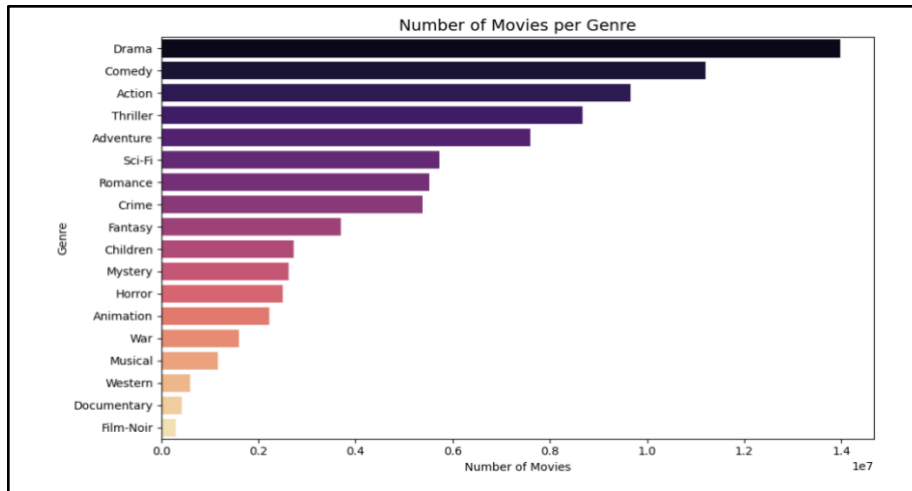
Aggregating ratings by title revealed **“The Shawshank Redemption (1994)”** as the most-rated film (~100k ratings), followed by **“Forrest Gump”** and **“Pulp Fiction”**, both from 1994. The tenth-most-reviewed film, **“The Fellowship of the Ring (2001)”**, had ~75k ratings. These results show a strong dominance of widely beloved classics.



3.4 Content Landscape: Genre Analysis

Genre counts show **Drama** as the most prevalent genre, followed by **Comedy** and **Action**, with **Film-Noir** and **Western** among the least common.

Violin plots of rating distributions revealed a key insight: **nearly all genres share the same rating pattern**, with medians centered at **4.0**. This suggests rating behavior is largely genre-independent, simplifying modeling since a single SVD model can generalize across genres without genre-specific adjustments.

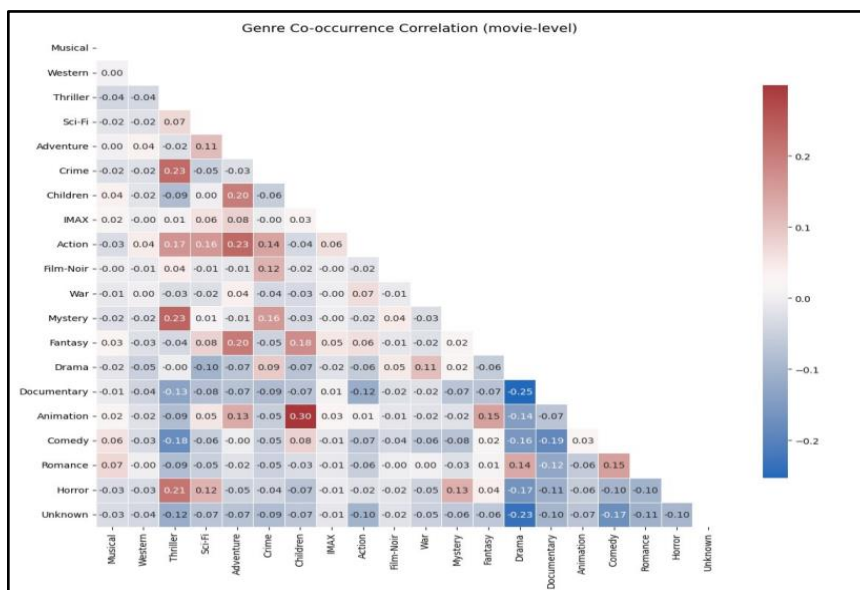


3.5 Genre Co-occurrence and Relationships

The dataset includes many movies tagged with multiple genres. A correlation heatmap was computed to analyze which genres are most likely to co-occur on the same film.¹ This analysis revealed several intuitive relationships and structural patterns in the movie-feature matrix:

- **Positive Correlations:** The strongest positive correlations were found between (Children, Animation) at +0.30 and (Crime, Thriller) at +0.23. (Comedy, Romance) also showed a positive correlation of +0.15.¹ These pairs represent common and expected genre combinations.
- **Negative Correlation:** A notable negative correlation of -0.25 was found between (Drama, Documentary), indicating that these two genres rarely overlap.¹
- **Weak/No Correlation:** Most other genre pairs showed weak or near-zero correlation, suggesting that while some mixing occurs, most genres are relatively distinct.¹

These correlations are not merely trivia; they reflect the underlying structure of the data that the SVD model will implicitly learn and leverage within its latent factors.



4.0 SVD Model Development and Performance

4.1 Model Selection and Validation Strategy

An 80/20 train-test split was used for reliable large-scale validation.

The `cleaned_movies.csv` dataset (32M ratings) was loaded into a Surprise `Dataset` and split with `test_size=0.2, random_state=42`, producing:

- **Trainset:** ~25.6M ratings
- **Testset:** ~6.4M ratings

4.2 Final Model Architecture and Training

The final model used Surprise's **SVD** algorithm. The final SVD model was trained using **50 latent factors** and **10 epochs**, with a **learning rate of 0.005** and **regularization set to 0.04**. A **random_state of 42** was used to ensure reproducible results.

4.3 Model Performance I: Accuracy (Error) Metrics

After training, the SVD model was applied to the 20% unseen testset to generate predictions for all 6.4 million user-item pairs. The model's predictive accuracy was then evaluated using the two standard error metrics for regression-based recommendation tasks:

- **RMSE (Root Mean Square Error): 0.8259**
- **MAE (Mean Absolute Error): 0.6243**

An RMSE of 0.8259 is a strong result, indicating that, on average, the model's rating predictions are off by approximately 0.826 stars on the 5-star scale. The MAE of 0.6243 provides a complementary view, showing the average absolute difference between the predicted and true ratings. This confirms the model's ability to accurately predict user ratings.

4.4 Model Performance II: Ranking (Top-K) Metrics

The model's Top-10 predictions for each user were then compared against this "ground truth" set of relevant items to calculate precision and recall:

- **Precision@10: 0.6625**
- **Recall@10: 0.6679**

A Precision@10 of 0.6625 is an excellent score. It signifies that, on average, 66.25% of the movies presented in a user's Top-10 recommendation list were items they would have genuinely considered "relevant" (i.e., would have rated 3.5 stars or higher). These metrics confirm that the model is not only accurate in its specific predictions but also highly effective at its core task: identifying and ranking items the user will value.

4.5 Qualitative Validation: Example Recommendations

For a sample user (ID 10), the model generated predicted ratings for unseen movies, ranging from **3.14** to **2.63**. A fast Top-N recommendation function was also implemented, handling both regular users and cold-start cases by falling back to popular titles. This serves as the deployable recommendation component.

5 - Supplemental Analyses and Modeling Diagnostics

5.1 Diagnostic: Model Behavior vs. Epochs (Learning Curve)

A learning-curve experiment on a **1% sample** showed that RMSE improves rapidly up to ~30 epochs, then plateaus. Gains beyond 30 epochs were minimal. This indicates that moderate values like **10–20 epochs** capture most of the benefit without unnecessary computation. RMSE dropped from **0.8872 at 5 epochs** to **0.8760 at 10 epochs**, **0.8600 at 20 epochs**, and **0.8509 at 30 epochs**. Beyond this point, gains were minimal, with **40 epochs** achieving only a slight improvement to **0.8490**. Training time increased gradually as well, rising from **14.7 seconds at 5 epochs** to **3.3 seconds at 40 epochs**, confirming that most performance benefits are captured within the first 20–30 epochs.

5.2 Diagnostic: Hyperparameter Tuning Experiments

To explore the parameter space and search for optimal SVD configurations, two separate hyperparameter tuning experiments were conducted on subsets of the data.¹

1. GridSearchCV (on 5% Sample):

A GridSearchCV was performed on a 5% sample of the dataset. This exhaustive search tested 24 different parameter combinations (`n_factors`: , `n_epochs`: , `lr_all`: [0.002, 0.005], `reg_all`: [0.02, 0.05]).

- **Best RMSE:** 0.8994
- **Best Parameters:** {'n_factors': 50, 'n_epochs': 20, 'lr_all': 0.005, 'reg_all': 0.05}

2. RandomizedSearch (on 10% Sample):

A RandomizedSearch (10 trials) was performed on a larger 10% sample of the dataset, exploring a similar parameter space.¹

- **Best RMSE:** 0.8955
- **Best Parameters:** {'n_factors': 100, 'n_epochs': 10, 'lr_all': 0.01, 'reg_all': 0.04}

These tuning experiments provided a baseline understanding of the parameter space, but their most valuable contribution comes when compared to the final model. The tuning runs on 5-10% of the data found optimal RMSEs in the ~0.895-0.900 range.¹ The final model (Section 4.0), which used a simple, non-tuned set of parameters (`n_factors`=50, `n_epochs`=10), achieved a *significantly* better RMSE of 0.8259. Overall, SVD behaved as a **data-hungry model**, with performance scaling strongly with larger training sets.

6.0 Conclusion

The project began with a meticulous preprocessing and feature engineering phase, transforming the 32-million-entry ml-32m dataset into a clean, structured, and feature-rich analytical file. The subsequent Exploratory Data Analysis provided critical, non-obvious context, revealing a user base with a strong positive rating bias and, most importantly, a highly stable collective taste profile that has remained consistent for decades.

The modeling phase successfully overcame a significant technical challenge related to large-scale cross-validation, leading to the adoption of a robust 80/20 train-test split for final validation.

The final SVD model is both accurate and effective. It demonstrated a strong predictive accuracy with an RMSE of 0.8259. Critically, it also excelled at its core business task: ranking. With a Precision@10 of 0.6625, the model proves it is highly capable of identifying and recommending items that users will find relevant and enjoyable. Supplemental analyses further revealed that the SVD algorithm's performance is profoundly-linked to data volume, with the final model trained on 80% of the data (25.6 million ratings) dramatically outperforming all hyperparameter tuning experiments conducted on smaller 5-10% subsets.

Group 8 Members

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- Contributed to data preprocessing and report preparation.

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